

Comprehensive Medical Physiology & Histology

This document provides a structured, in-depth study guide covering the fundamental pillars of human biology for medical and allied health students. From the microscopic architecture of tissues to the integrated physiology of organ systems, each topic is presented with clear explanations, visual diagrams, and clinical correlations. The content spans six major domains: Basic Histology, Neurophysiology, Muscle & Sensory Physiology, Digestive & Respiratory Physiology, Excretory & Circulatory Physiology, and Reproductive Histology. Use this as a reference for exam preparation, clinical correlation, and foundational understanding of human biology.

[HISTOLOGY](#)[NEUROPHYSIOLOGY](#)[MUSCLE PHYSIOLOGY](#)[SENSORY SYSTEMS](#)[ORGAN SYSTEMS](#)

Basic Histology: Epithelial & Connective Tissue

Histology is the microscopic study of tissues — the building blocks of organs. The four primary tissue types are epithelial, connective, muscular, and nervous tissue. Each has distinct structural and functional characteristics that enable the body to maintain homeostasis, respond to stimuli, and execute complex physiological processes. Understanding tissue architecture is foundational for pathology, surgery, and clinical diagnostics.

Epithelial Tissue

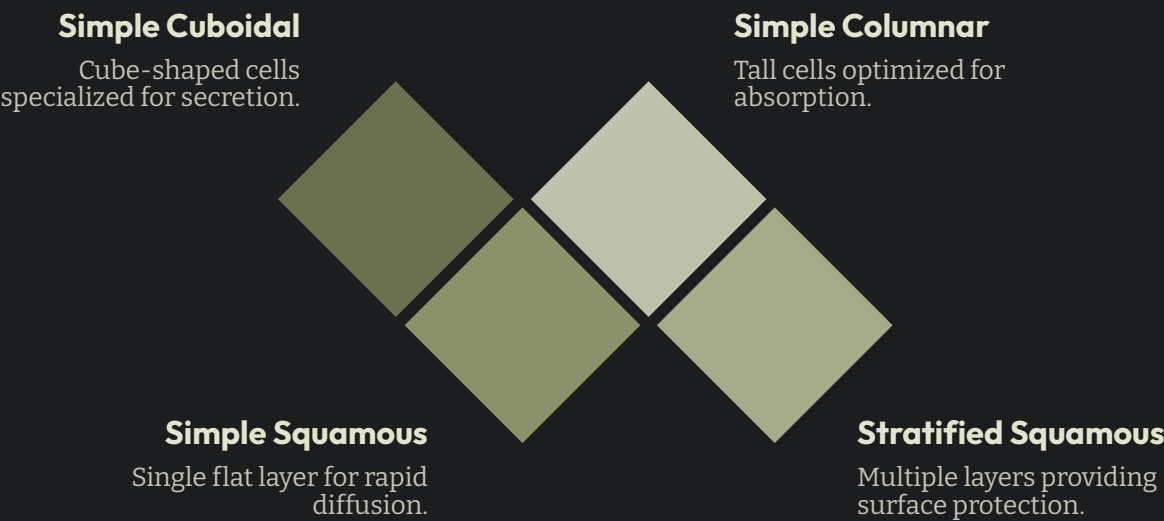
Epithelial tissue covers body surfaces, lines cavities, and forms glands. It is characterized by tightly packed cells with minimal intercellular material, a basement membrane, polarity (apical and basal surfaces), and avascularity. Epithelial classification is based on cell layers (simple vs. stratified) and cell shape (squamous, cuboidal, columnar). Specializations include microvilli (brush border), cilia, and keratinization. Functions include protection, absorption, secretion, filtration, and sensation. The apical surface may bear cilia in respiratory epithelium or microvilli in intestinal epithelium to increase surface area for absorption. Goblet cells interspersed within columnar epithelium secrete mucus for lubrication and protection.

Simple Squamous Single layer of flat cells. Found in alveoli, blood vessels (endothelium), and serous membranes. Functions in diffusion and filtration.	Simple Cuboidal Single layer of cube-shaped cells. Found in kidney tubules, thyroid follicles, and gland ducts. Functions in secretion and absorption.
Simple Columnar Single layer of tall cells. Lines the stomach and intestines. Often contains goblet cells. Functions in absorption and secretion.	Stratified Squamous Multiple layers; surface cells are flat. Non-keratinized in oral cavity and esophagus; keratinized in skin (epidermis). Provides protection.
Pseudostratified Ciliated All cells touch basement membrane but not all reach surface. Ciliated type lines respiratory tract. Contains goblet cells for mucus secretion.	Transitional Epithelium Stratified epithelium that changes shape with stretching. Found exclusively in the urinary bladder and ureters. Allows distension.

Connective Tissue

Connective tissue is the most abundant and widely distributed tissue in the body. It consists of cells, fibers (collagen, elastic, reticular), and ground substance (matrix). Functions include support, binding, protection, insulation, and transportation. Connective tissue proper includes loose and dense varieties. Specialized connective tissues include cartilage, bone, blood, and adipose tissue. Collagen fibers provide tensile strength, elastic fibers allow recoil, and reticular fibers form delicate supporting networks in lymphoid organs.

Connective Tissue Proper • Loose CT: Areolar (most common, beneath epithelia), Adipose (fat storage, insulation), Reticular (lymphoid organs) • Dense Regular CT: Parallel collagen fibers; tendons and ligaments • Dense Irregular CT: Randomly arranged fibers; dermis, organ capsules • Elastic CT: Abundant elastic fibers; walls of large arteries	Specialized Connective Tissues • Cartilage: Hyaline (most common, at joint surfaces), Elastic (external ear, epiglottis), Fibrocartilage (intervertebral discs) • Bone: Compact (dense outer layer) and Spongy (trabecular, inner) • Blood: Fluid matrix (plasma); transports gases, nutrients, wastes • Adipose: Energy storage, thermal insulation, endocrine function
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The diagram above illustrates the structural progression from the thinnest, most permeable epithelium (simple squamous) optimized for diffusion, through secretory and absorptive forms, to the multi-layered stratified squamous epithelium designed for mechanical protection against abrasion.

Basic Histology: Muscular, Nervous Tissue, Bone, Cartilage & Ossification

Muscular Tissue

Muscular tissue is specialized for contraction and is responsible for movement, posture, and heat production. There are three types: skeletal (striated, voluntary, multinucleated), cardiac (striated, involuntary, branched, intercalated discs), and smooth (non-striated, involuntary, spindle-shaped). Skeletal muscle fibers are organized into fascicles surrounded by perimysium; each fiber is wrapped in endomysium. The entire muscle is encased in epimysium. Cardiac muscle contains intercalated discs with gap junctions for electrical coupling and desmosomes for mechanical strength. Smooth muscle lines hollow organs and blood vessels and contracts via calmodulin-mediated pathways rather than troponin.

Skeletal Muscle

Voluntary, striated, multinucleated. Attached to bones via tendons. Controlled by somatic nervous system. Shows cross-striations due to sarcomere organization.

Cardiac Muscle

Involuntary, striated, single nucleus. Found only in the heart. Intercalated discs with gap junctions allow synchronized contraction. Autorhythmic.

Smooth Muscle

Involuntary, non-striated, spindle-shaped. Lines hollow organs (stomach, bladder, blood vessels). Contracts slowly but sustains contraction for long periods.

Nervous Tissue

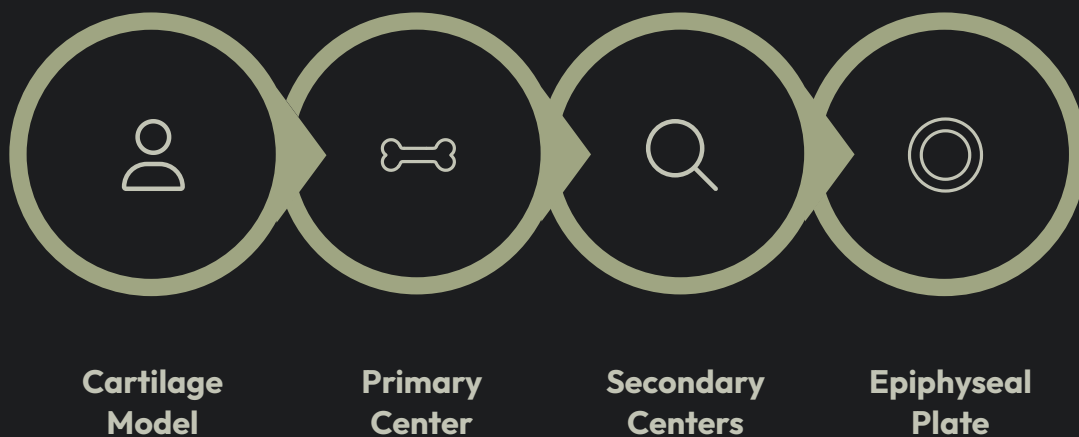
Nervous tissue is the primary component of the nervous system and is specialized for receiving stimuli, conducting impulses, and integrating information. It consists of two cell types: neurons (functional units that transmit electrical signals) and neuroglia (supporting cells). In the CNS, neuroglia include astrocytes, oligodendrocytes, microglia, and ependymal cells. In the PNS, Schwann cells produce myelin and satellite cells provide support. Neurons are classified structurally as multipolar, bipolar, or unipolar, and functionally as sensory (afferent), motor (efferent), or interneurons. The axon hillock is the site of action potential initiation, and myelination by oligodendrocytes (CNS) or Schwann cells (PNS) dramatically increases conduction velocity through saltatory conduction at nodes of Ranvier.

Bone & Cartilage

Bone is a specialized connective tissue with a mineralized matrix (hydroxyapatite crystals) that provides rigidity and strength. Compact bone is organized into osteons (Haversian systems) with concentric lamellae surrounding a central canal containing blood vessels. Spongy bone contains trabeculae arranged along lines of stress. Bone cells include osteoblasts (bone-forming), osteocytes (mature, maintain bone), and osteoclasts (bone-resorbing, multinucleated). Cartilage is avascular and consists of chondrocytes in lacunae within a firm gel-like matrix. Hyaline cartilage is the most common type and serves as a model for endochondral ossification. The perichondrium (absent in articular cartilage) provides nutrients and contains chondrogenic cells for growth and repair.

Ossification

Ossification is the process of bone formation. There are two types: intramembranous ossification (direct formation from mesenchymal tissue; forms flat bones of the skull and clavicle) and endochondral ossification (bone replaces a hyaline cartilage model; forms most bones of the body). In endochondral ossification, a primary ossification center forms in the diaphysis during fetal development, and secondary centers form in the epiphyses after birth. The epiphyseal plate (growth plate) allows longitudinal growth until fusion in early adulthood. Bone remodeling is a continuous process involving balanced osteoblast and osteoclast activity regulated by hormones (PTH, calcitonin, vitamin D) and mechanical stress. Wolff's law states that bone adapts to the loads under which it is placed.



Endochondral ossification is the predominant mechanism by which the skeleton forms. The cartilage model is progressively replaced by bone tissue, beginning at the primary ossification center in the diaphysis and extending toward both epiphyses. The epiphyseal plate remains active during childhood and adolescence, permitting longitudinal growth until hormonal changes trigger its closure in early adulthood.

Neurophysiology: Neuron Structure, Membrane Potential & Action Potential

Neurophysiology is the study of nervous system function at the cellular and molecular level. Understanding how neurons generate and transmit electrical signals is fundamental to comprehending all nervous system activity, from reflexes to cognition. The electrical properties of neurons depend on ion gradients, selective membrane permeability, and specialized protein channels.

Neuron Structure

The neuron is the structural and functional unit of the nervous system. A typical neuron consists of a cell body (soma) containing the nucleus and organelles, dendrites that receive incoming signals, and a single axon that conducts impulses away from the soma. The axon hillock is the site of action potential initiation. Axons may be myelinated (by Schwann cells in PNS or oligodendrocytes in CNS) or unmyelinated. Myelinated axons conduct impulses via saltatory conduction at nodes of Ranvier, greatly increasing velocity. Axon terminals contain synaptic vesicles filled with neurotransmitters. Neurons are classified as multipolar (most common, motor neurons), bipolar (sensory neurons of retina and olfactory epithelium), or unipolar/pseudounipolar (dorsal root ganglia). Nissl bodies (rough ER) are prominent in the soma and dendrites but absent from the axon hillock and axon.



Dendrites

Receive incoming signals from other neurons or receptors. Highly branched to maximize surface area. Contain receptor proteins and ligand-gated ion channels.



Axon Hillock

Cone-shaped region where axon originates. Has the lowest threshold for action potential initiation. Summation of graded potentials occurs here.



Cell Body (Soma)

Contains nucleus, Nissl bodies (rough ER), and Golgi apparatus. Integrates incoming signals. Metabolic center of the neuron.



Myelin Sheath

Lipid-rich insulation formed by Schwann cells (PNS) or oligodendrocytes (CNS). Increases conduction velocity via saltatory conduction at nodes of Ranvier.

Resting Membrane Potential

The resting membrane potential (RMP) is the electrical potential difference across the neuronal membrane at rest, typically **-70 mV** (inside negative relative to outside). This potential is maintained by: (1) the Na^+/K^+ ATPase pump, which actively transports 3 Na^+ out and 2 K^+ in, creating ionic gradients; (2) selective membrane permeability — at rest, the membrane is $\sim 100\times$ more permeable to K^+ than Na^+ due to open K^+ leak channels; (3) the presence of large negatively charged intracellular proteins (anions) that cannot cross the membrane. The equilibrium potential for K^+ is approximately -90 mV, and for Na^+ it is approximately +60 mV. The RMP is closer to the K^+ equilibrium potential because K^+ permeability dominates at rest. The Goldman-Hodgkin-Katz equation accounts for the relative permeabilities of multiple ions to calculate the actual membrane potential.

Action Potential

An action potential (AP) is a rapid, all-or-none electrical event that propagates along the axon. It consists of four phases: (1) **Depolarization** — voltage-gated Na^+ channels open when threshold (-55 mV) is reached; Na^+ rushes in, membrane potential rises to +30 mV; (2) **Repolarization** — Na^+ channels inactivate, voltage-gated K^+ channels open; K^+ flows out, membrane potential returns toward resting level; (3) **Hyperpolarization** — K^+ channels remain open slightly longer, causing membrane potential to briefly drop below -70 mV; (4) **Refractory periods** — absolute refractory period (no new AP possible, Na^+ channels inactivated) and relative refractory period (stronger stimulus required). AP propagation is unidirectional due to the refractory period. In myelinated axons, APs "jump" between nodes of Ranvier (saltatory conduction), increasing speed up to 120 m/s.



The action potential is an all-or-none event — once threshold is reached, the full depolarization-repolarization sequence occurs regardless of stimulus strength. The refractory periods ensure unidirectional propagation and limit the maximum firing frequency of the neuron, which is critical for encoding information in the nervous system.

Neurophysiology: Synapse, Neuromuscular Junction & Reflex Arc

Synapse

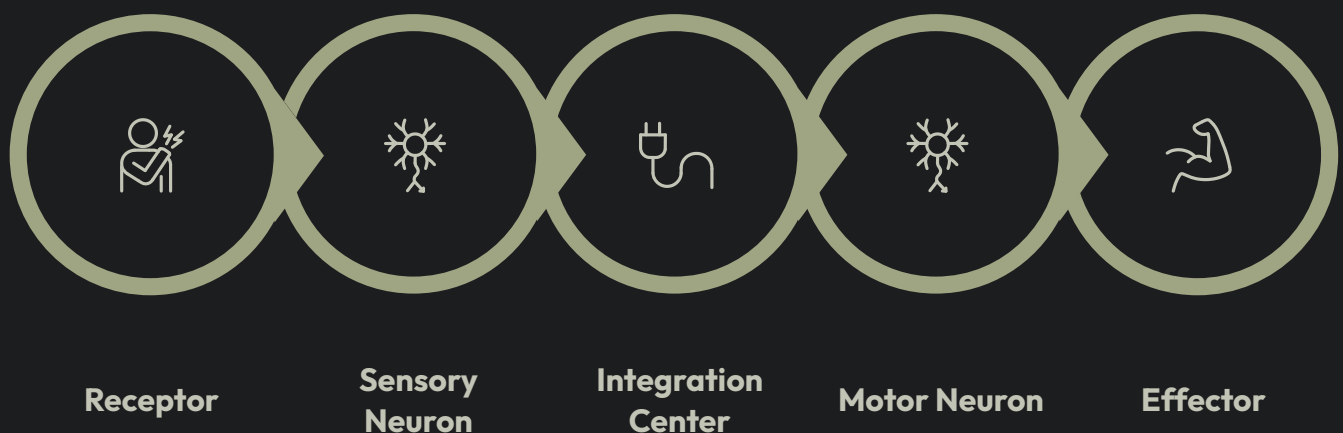
A synapse is the specialized junction between two neurons (or between a neuron and an effector cell) where nerve impulses are transmitted. Synapses are classified as **electrical** (gap junctions allow direct ion flow; fast, bidirectional) or **chemical** (neurotransmitter-mediated; unidirectional, modifiable). Chemical synapses are far more common in the human nervous system. The presynaptic terminal contains synaptic vesicles filled with neurotransmitter. When an action potential arrives, voltage-gated Ca^{2+} channels open, Ca^{2+} enters, and vesicles fuse with the presynaptic membrane (exocytosis), releasing neurotransmitter into the synaptic cleft (~20–50 nm wide). The neurotransmitter binds to receptors on the postsynaptic membrane, causing either depolarization (excitatory postsynaptic potential, EPSP) or hyperpolarization (inhibitory postsynaptic potential, IPSP). Common neurotransmitters include acetylcholine, glutamate (excitatory), GABA, glycine (inhibitory), dopamine, serotonin, and norepinephrine. Neurotransmitter action is terminated by enzymatic degradation (e.g., acetylcholinesterase breaks down ACh), reuptake into the presynaptic terminal, or diffusion away from the cleft. Spatial and temporal summation of EPSPs and IPSPs at the axon hillock determines whether threshold is reached and an action potential is generated.

Neuromuscular Junction

The neuromuscular junction (NMJ) is the specialized chemical synapse between a motor neuron and a skeletal muscle fiber. The presynaptic terminal (motor end plate) contains vesicles of **acetylcholine (ACh)**. The postsynaptic membrane (motor end plate) is highly folded into junctional folds and contains nicotinic ACh receptors (ligand-gated Na^+/K^+ channels). When an action potential arrives at the motor neuron terminal, Ca^{2+} influx triggers ACh release. ACh binds to receptors, causing Na^+ influx and depolarization (end plate potential, EPP). The EPP always exceeds threshold and triggers a muscle action potential that propagates along the sarcolemma and into T-tubules, initiating excitation-contraction coupling. ACh is rapidly degraded by **acetylcholinesterase (AChE)** in the synaptic cleft, terminating the signal. Myasthenia gravis is an autoimmune disease where antibodies block ACh receptors, causing muscle weakness. Curare and succinylcholine are drugs that interfere with NMJ transmission.

Reflex Arc

A reflex arc is the simplest functional unit of the nervous system — a rapid, involuntary, stereotyped response to a specific stimulus. The basic components are: (1) **Receptor** — detects the stimulus; (2) **Sensory (afferent) neuron** — transmits impulse to CNS; (3) **Integration center** — one or more synapses in CNS (spinal cord or brainstem); (4) **Motor (efferent) neuron** — transmits impulse from CNS to effector; (5) **Effector** — muscle or gland that responds. Reflexes are classified as **monosynaptic** (single synapse, e.g., patellar reflex) or **polysynaptic** (multiple synapses with interneurons, e.g., withdrawal reflex). The stretch reflex (myotatic reflex) is monosynaptic and involves muscle spindles detecting muscle length. The Golgi tendon reflex is polysynaptic and protective, preventing excessive tension. Reciprocal inhibition ensures that when an agonist muscle contracts, the antagonist relaxes. Reflex testing is a key component of the neurological examination.



The reflex arc represents the most direct neural pathway in the body. Because integration occurs at the spinal cord level, reflex responses occur before the brain consciously perceives the stimulus — a critical protective mechanism. The patellar reflex is a classic example of a monosynaptic stretch reflex, while the withdrawal reflex from a painful stimulus involves polysynaptic pathways with reciprocal inhibition of antagonistic muscles.

Muscle & Sensory Physiology

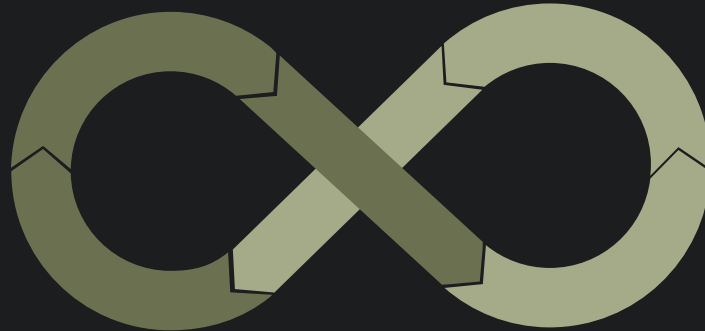
The physiology of muscle contraction and sensory perception represents two of the most elegantly coordinated systems in human biology. Skeletal muscle converts chemical energy into mechanical work through the sliding filament mechanism, while sensory organs transduce environmental stimuli into neural signals that the brain interprets as vision, hearing, and other sensations.

Skeletal Muscle Contraction

Skeletal muscle contraction is explained by the **sliding filament theory**. The functional unit is the **sarcomere**, bounded by Z-discs and containing thick filaments (myosin) and thin filaments (actin, tropomyosin, troponin). At rest, tropomyosin blocks myosin-binding sites on actin. When an action potential arrives via the motor neuron, it propagates along the sarcolemma and T-tubules, triggering Ca^{2+} release from the sarcoplasmic reticulum. Ca^{2+} binds to troponin C, causing a conformational change that moves tropomyosin and exposes myosin-binding sites on actin. Myosin heads (cross-bridges) bind to actin, perform a power stroke (pulling actin toward the center of the sarcomere), detach upon ATP binding, and reset (cocking) upon ATP hydrolysis. This cross-bridge cycling continues as long as Ca^{2+} and ATP are available. The sarcomere shortens, but filament length remains constant. Relaxation occurs when Ca^{2+} is pumped back into the sarcoplasmic reticulum by Ca^{2+} -ATPase, and tropomyosin re-blocks binding sites.

Cross-Bridge Formation

Power Stroke



Myosin Head Reactivation

Cross-Bridge Detachment

The cross-bridge cycle is the molecular basis of muscle contraction. Each cycle consumes one ATP molecule and produces a small displacement of the actin filament. Millions of cross-bridges cycling asynchronously produce smooth, sustained contraction. The frequency of cycling determines the speed of shortening, while the number of active cross-bridges determines force production.

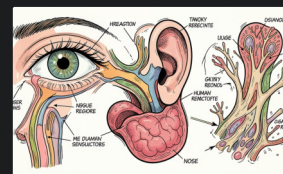
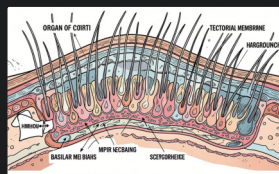
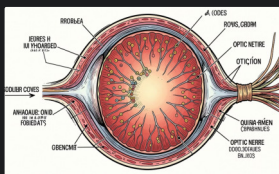
Muscle Twitch & Tetanus

A **muscle twitch** is the mechanical response of a muscle fiber to a single action potential. It has three phases: (1) **Latent period** (1–2 ms) — AP propagates, Ca^{2+} released, no tension yet; (2) **Contraction phase** (10–100 ms) — cross-bridge cycling, tension increases; (3) **Relaxation phase** (10–100 ms) — Ca^{2+} reabsorbed, tension decreases. A single twitch produces submaximal force. When stimuli are delivered at increasing frequency, twitches summate (**temporal summation**). At moderate frequency, unfused (incomplete) tetanus occurs with wavering tension. At high frequency, fused (complete) tetanus occurs — a sustained, maximal contraction without relaxation. Tetanus produces 3–4× more force than a single twitch. The **length-tension relationship** shows that maximal force is generated at optimal sarcomere length (2.0–2.2 μm), where actin-myosin overlap is ideal. The **force-velocity relationship** shows that as load increases, shortening velocity decreases. Motor unit recruitment (size principle) and frequency summation are the two mechanisms by which the nervous system gradates muscle force.

Physiology of Vision & Hearing

Vision begins when light enters the eye through the cornea, passes through the aqueous humor, pupil, and lens (which focuses light by accommodation), and reaches the retina. The retina contains photoreceptors: **rods** (120 million, peripheral retina, high sensitivity, low acuity, scotopic/night vision, contain rhodopsin) and **cones** (6–7 million, concentrated in fovea centralis, color vision, high acuity, photopic/day vision; three types: S-cones for blue, M-cones for green, L-cones for red). Phototransduction involves the conversion of light energy into electrical signals. In darkness, photoreceptors are depolarized and release glutamate continuously. Light activates rhodopsin (in rods) or photopsins (in cones), activating transducin (G-protein), which activates phosphodiesterase, reducing cGMP levels, closing Na^+ channels, and causing hyperpolarization. This hyperpolarization reduces glutamate release, signaling bipolar cells and ultimately ganglion cells whose axons form the optic nerve. The visual pathway continues through the optic chiasm (partial decussation), optic tract, lateral geniculate nucleus (LGN) of the thalamus, and primary visual cortex (V1) in the occipital lobe.

Hearing involves sound waves entering the external ear, vibrating the tympanic membrane, and transmitting vibrations through the ossicles (malleus, incus, stapes) to the oval window of the cochlea. Fluid waves in the cochlea cause displacement of the basilar membrane. High-frequency sounds maximally displace the base; low-frequency sounds displace the apex (tonotopic organization). Hair cells in the organ of Corti are the sensory receptors: inner hair cells (one row, primary sensory transducers) and outer hair cells (three rows, amplify and fine-tune the signal). Stereocilia deflection opens mechanically gated K^+ channels (endolymph has high K^+ concentration), causing depolarization, Ca^{2+} influx, and glutamate release onto afferent nerve fibers of the spiral ganglion. Signals travel via the cochlear nerve to the cochlear nuclei, superior olivary complex (sound localization), inferior colliculus, medial geniculate nucleus, and primary auditory cortex (temporal lobe). The vestibular system (semicircular canals and otolith organs) detects head rotation and linear acceleration for balance.



Visual Receptors

Rods (scotopic vision, peripheral) and cones (photopic and color vision, foveal). Three cone types detect red, green, and blue wavelengths. Phototransduction converts light to neural signals via rhodopsin and photopsins.

Auditory Receptors

Inner hair cells (sensory transduction) and outer hair cells (amplification) in the organ of Corti. Tonotopic organization: base detects high frequencies, apex detects low frequencies. Endolymph's high K^+ concentration is essential for mechanotransduction.

Sense Organs Overview

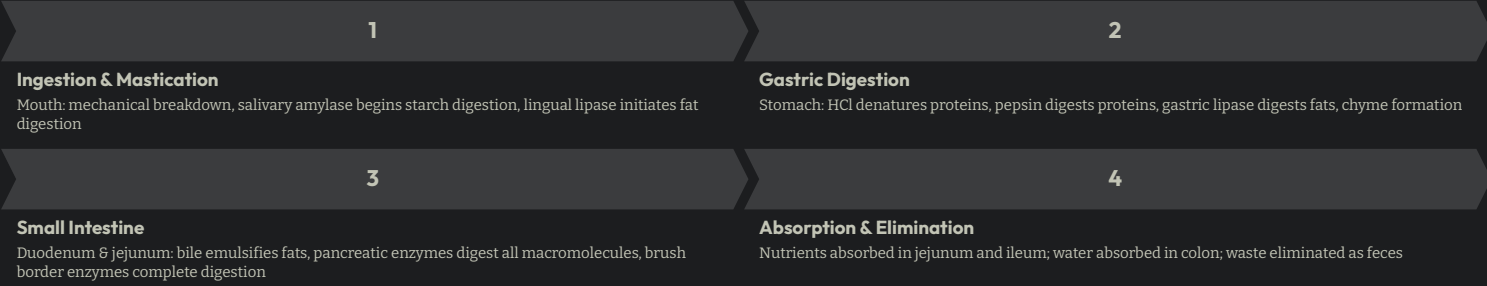
The five special senses (vision, hearing, taste, smell, equilibrium) and general somatic senses (touch, pain, temperature, proprioception) use specialized receptors to transduce specific stimulus modalities into neural signals.

Digestive & Respiratory Physiology

The digestive and respiratory systems work in concert to provide the body with nutrients and oxygen while eliminating metabolic waste products. Their physiology involves complex mechanical and chemical processes, hormonal regulation, and gas exchange mechanisms that are essential for cellular metabolism and energy production.

Digestive System

Digestion involves the mechanical and chemical breakdown of food into absorbable molecules. The process begins in the mouth with mastication and salivary amylase (ptyalin) initiating starch digestion. The stomach performs mechanical churning and secretes HCl (from parietal cells, pH ~2) and pepsinogen (from chief cells), which is converted to pepsin for protein digestion. Gastrin stimulates acid secretion. The small intestine is the primary site of digestion and absorption. The duodenum receives bile from the liver/gallbladder (emulsifies fats) and pancreatic juice containing amylase, lipase, trypsinogen, chymotrypsinogen, and bicarbonate. Brush border enzymes (maltase, lactase, peptidases) complete digestion. Absorption occurs primarily in the jejunum: carbohydrates and proteins via active transport and facilitated diffusion; fats via micelle formation and chylomicron assembly into lacteals. The large intestine absorbs water and electrolytes, and houses gut microbiota that synthesize vitamin K and B vitamins. Peristalsis and segmentation are the primary motility patterns. The enteric nervous system (myenteric and submucosal plexuses) coordinates digestive activity independently, though it is modulated by the autonomic nervous system.



Respiratory System

The respiratory system facilitates gas exchange between the atmosphere and blood. Ventilation (breathing) moves air in and out of the lungs. Inspiration is active (diaphragm and external intercostals contract); expiration is passive at rest (elastic recoil). Lung volumes include tidal volume (500 mL), inspiratory reserve volume, expiratory reserve volume, and residual volume. Vital capacity = IRV + TV + ERV. Total lung capacity = VC + RV. Gas exchange occurs across the respiratory membrane (alveolar-capillary membrane, ~0.5 µm thick) by simple diffusion, governed by Fick's law. O₂ diffuses from alveoli (PO₂ ~104 mmHg) to pulmonary capillaries (PO₂ ~40 mmHg); CO₂ diffuses in the opposite direction. Despite CO₂ having a smaller partial pressure gradient, it diffuses 20× faster than O₂ due to higher solubility. Oxygen is transported in blood: 98.5% bound to hemoglobin (Hb), 1.5% dissolved in plasma. CO₂ is transported as bicarbonate (70%), carbamino compounds (23%), and dissolved (7%). The chloride shift (Hamburger phenomenon) maintains electrical neutrality as HCO₃⁻ moves out of RBCs in exchange for Cl⁻. Respiratory control centers in the medulla (dorsal and ventral respiratory groups) and pons (pneumotaxic and apneustic centers) regulate breathing rate and depth. Central chemoreceptors (medulla) respond to CO₂/pH in cerebrospinal fluid; peripheral chemoreceptors (carotid and aortic bodies) respond to low PO₂, high PCO₂, and low pH.

Respiratory Pigments & Dissociation Curves

Hemoglobin (Hb) is the primary respiratory pigment in humans — a tetrameric protein (2 α + 2 β chains) with four heme groups, each binding one O₂ molecule. The **oxygen-hemoglobin dissociation curve** is S-shaped (sigmoid), reflecting cooperative binding: binding of the first O₂ molecule increases affinity for subsequent molecules. At alveolar PO₂ (100 mmHg), Hb is ~97.5% saturated; at tissue PO₂ (40 mmHg), saturation drops to ~75%, releasing O₂ to tissues. The **Bohr effect**: increased PCO₂, decreased pH, and increased temperature shift the curve rightward, decreasing Hb-O₂ affinity and enhancing O₂ unloading in metabolically active tissues. **2,3-BPG** (bisphosphoglycerate) also shifts the curve rightward by binding to deoxyhemoglobin. Fetal hemoglobin (HbF, α₂γ₂) has higher O₂ affinity than adult Hb (HbA), facilitating O₂ transfer across the placenta. The **CO₂ dissociation curve** is more linear; increased PO₂ shifts it leftward (Haldane effect — deoxygenated Hb carries more CO₂).

Carbon Monoxide Poisoning

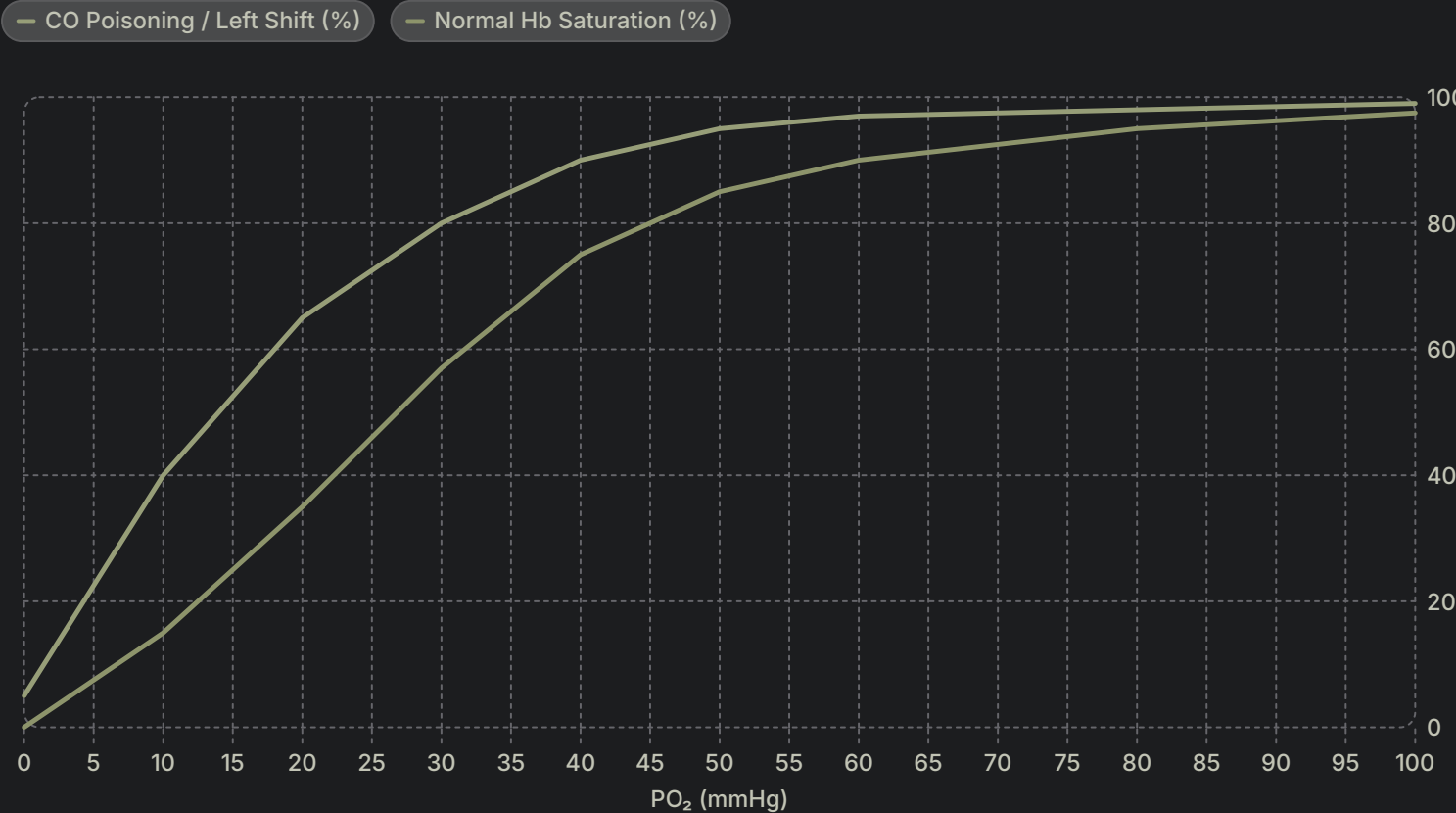
Carbon monoxide (CO) is a colorless, odorless gas produced by incomplete combustion. CO binds to hemoglobin with an affinity **200–250× greater than O₂**, forming carboxyhemoglobin (COHb). This has two devastating effects: (1) it reduces the O₂-carrying capacity of blood by occupying Hb binding sites; (2) it shifts the O₂-Hb dissociation curve leftward, increasing the affinity of remaining heme groups for O₂ and impairing O₂ release to tissues. Even low CO concentrations (0.1%) can saturate 50% of Hb. Symptoms include headache, dizziness, nausea, confusion, and at high levels, unconsciousness and death. Tissues with high O₂ demand (brain, heart) are most vulnerable. Treatment involves 100% O₂ therapy (reduces COHb half-life from 4–5 hours to ~80 minutes) or hyperbaric oxygen therapy for severe cases. CO also binds to myoglobin and mitochondrial cytochrome oxidase, further impairing cellular respiration.

Oxygen-Hemoglobin Dissociation Curve

- **Sigmoid shape** — cooperative binding
- **Plateau** at PO₂ >60 mmHg — safe margin for alveolar hypoxia
- **Steep portion** at PO₂ 20–40 mmHg — facilitates O₂ unloading
- **Right shift** (Bohr effect): ↑PCO₂, ↓pH, ↑temperature, ↑2,3-BPG
- **Left shift**: ↓PCO₂, ↑pH, ↓temperature, HbF, CO poisoning

Factors Affecting Gas Transport

- **Bohr Effect**: Acidic, high-CO₂ environment in tissues promotes O₂ release
- **Haldane Effect**: Deoxygenated Hb carries more CO₂ than oxygenated Hb
- **Chloride Shift**: HCO₃⁻/Cl⁻ exchange maintains electrical neutrality in RBCs
- **CO Poisoning**: CO binds Hb 200× more avidly than O₂, shifting curve leftward
- **High Altitude**: Lower atmospheric PO₂ reduces Hb saturation; compensated by ↑2,3-BPG and erythropoiesis



The oxygen-hemoglobin dissociation curve demonstrates the sigmoid relationship between PO₂ and Hb saturation. The normal curve (green) shows cooperative binding with a plateau above 60 mmHg. A left shift (red, as in CO poisoning or alkalosis) increases affinity but impairs tissue unloading. A right shift (blue, Bohr effect) decreases affinity, enhancing O₂ delivery to active tissues. The steep portion of the curve (PO₂ 20–40 mmHg) is where most O₂ unloading occurs in systemic capillaries.

Excretory & Circulatory Physiology: Renal Function & Blood

The kidneys and cardiovascular system are intimately linked in maintaining homeostasis. The kidneys filter blood, regulate fluid and electrolyte balance, maintain acid-base equilibrium, and produce hormones. The blood and clotting mechanisms ensure oxygen delivery and hemostasis. Together, these systems sustain the internal environment necessary for cellular function.

Renal Physiology & Urine Formation

The functional unit of the kidney is the **nephron** (~1 million per kidney). Each nephron consists of a renal corpuscle (glomerulus + Bowman's capsule) and a renal tubule (PCT, loop of Henle, DCT, collecting duct). Urine formation involves three processes:

1. Glomerular Filtration

Blood plasma is filtered across the glomerular capillary wall into Bowman's capsule. Filtration barrier: fenestrated endothelium → basement membrane → podocyte foot processes (slit diaphragm). GFR = ~125 mL/min (180 L/day). Driven by net filtration pressure (hydrostatic pressure minus oncotic pressure). Autoregulated by myogenic mechanism and tubuloglomerular feedback. Not filtered: blood cells, large proteins.

2. Tubular Reabsorption

Substances are reclaimed from filtrate back into peritubular capillaries. **PCT:** reabsorbs 65% of Na^+ and water, 100% of glucose and amino acids (via secondary active transport), 80% of HCO_3^- . **Loop of Henle:** descending limb permeable to water only (concentrates filtrate); ascending limb impermeable to water, actively transports Na^+ , K^+ , Cl^- (dilutes filtrate). Countercurrent multiplier creates medullary osmotic gradient (300–1200 mOsm/L). **DCT & Collecting Duct:** fine-tuning under hormonal control — aldosterone (Na^+ reabsorption, K^+ secretion), ADH/vasopressin (water reabsorption via aquaporins).

3. Tubular Secretion

Substances are actively transported from peritubular capillaries into tubular fluid. Includes H^+ (acid-base balance), K^+ (regulated by aldosterone), creatinine, drugs (penicillin), and toxins. Secretion is important for eliminating substances not fully filtered and for fine-tuning electrolyte and pH balance.

Acid-Base Balance

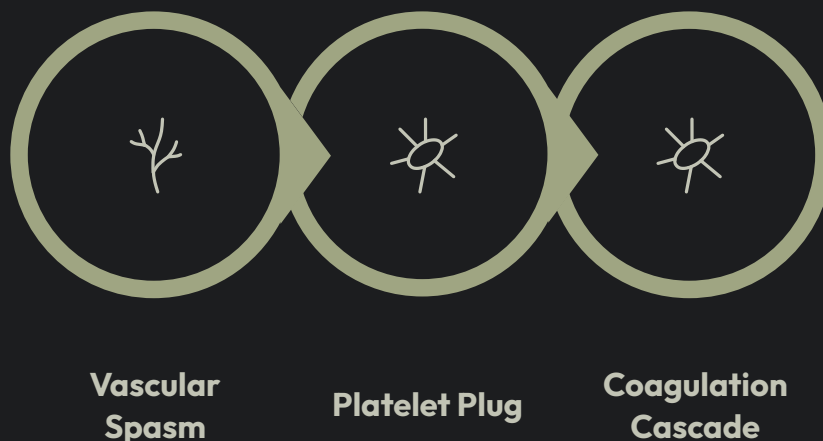
Normal arterial pH is 7.35–7.45. The body maintains pH through three mechanisms: (1) **Chemical buffers** (immediate) — bicarbonate buffer system (most important in ECF), phosphate buffer (important in urine and ICF), protein buffers (hemoglobin, plasma proteins); (2) **Respiratory compensation** (minutes to hours) — increased ventilation blows off CO_2 in acidosis; decreased ventilation retains CO_2 in alkalosis; (3) **Renal compensation** (hours to days) — kidneys excrete H^+ (as titratable acid and NH_4^+) and reabsorb/generate HCO_3^- . The kidneys are the ultimate regulators of acid-base balance. Metabolic acidosis ($\downarrow \text{HCO}_3^-$) is compensated by hyperventilation; metabolic alkalosis ($\uparrow \text{HCO}_3^-$) by hypoventilation. Respiratory acidosis ($\uparrow \text{PCO}_2$) is compensated by renal HCO_3^- retention; respiratory alkalosis ($\downarrow \text{PCO}_2$) by renal HCO_3^- excretion. The Henderson-Hasselbalch equation: $\text{pH} = 6.1 + \log\left(\frac{[\text{HCO}_3^-]}{0.03 \times \text{PCO}_2}\right)$.

Blood & Hemoglobin

Blood is a specialized connective tissue comprising **plasma** (55%; water, proteins, electrolytes, hormones, nutrients) and **formed elements** (45%): erythrocytes (RBCs, 4.5–5.5 million/ μL , carry O_2 and CO_2 , no nucleus, biconcave disc, lifespan 120 days, produced in red bone marrow under erythropoietin stimulation), leukocytes (WBCs, 5,000–10,000/ μL ; neutrophils 60%, lymphocytes 30%, monocytes 6%, eosinophils 3%, basophils 1%), and platelets (150,000–400,000/ μL , cell fragments from megakaryocytes, essential for hemostasis). **Hemoglobin** is a tetrameric protein ($2\alpha + 2\beta$ chains in adults, HbA) with four heme groups, each containing Fe^{2+} that binds one O_2 molecule. Normal Hb concentration: 13.5–17.5 g/dL (men), 12.0–15.5 g/dL (women). Hb also buffers H^+ and transports CO_2 as carbaminohemoglobin. ABO and Rh blood group systems are critical for transfusion compatibility.

Blood Clotting (Hemostasis)

Hemostasis prevents blood loss after vascular injury through three overlapping mechanisms: (1) **Vascular spasm** — immediate vasoconstriction reduces blood flow; (2) **Platelet plug formation** — platelets adhere to exposed collagen via von Willebrand factor, become activated, release ADP and thromboxane A_2 , and aggregate to form a temporary plug; (3) **Coagulation cascade** — a series of enzymatic reactions converting soluble fibrinogen to insoluble fibrin. The **extrinsic pathway** is initiated by tissue factor (Factor III) released from damaged tissue, activating Factor VII. The **intrinsic pathway** is initiated by contact activation (Factor XII on exposed collagen), involving Factors XII, XI, IX, and VIII. Both pathways converge at Factor X activation (common pathway), which with Factor V converts prothrombin (Factor II) to thrombin. Thrombin cleaves fibrinogen to fibrin monomers, which polymerize and are cross-linked by Factor XIII to form a stable clot. Vitamin K is essential for synthesis of Factors II, VII, IX, and X. Anticoagulants include antithrombin III, protein C, and protein S. Fibrinolysis (plasmin-mediated clot dissolution) restores vessel patency after healing.



Hemostasis is a tightly regulated process that must balance clot formation at sites of injury with the prevention of pathological thrombosis. The coagulation cascade amplifies the initial signal through a protease cascade, ultimately generating sufficient thrombin to form a stable fibrin clot. Deficiencies in clotting factors result in bleeding disorders (e.g., hemophilia A = Factor VIII deficiency), while excessive clotting leads to thrombosis and embolism.

Circulatory System, Evolution of Heart & Nervous System in Vertebrates

Circulatory System

The human cardiovascular system is a closed, double-circuit system comprising the heart, blood vessels, and blood. The **systemic circulation** delivers oxygenated blood from the left ventricle through the aorta to body tissues and returns deoxygenated blood via the venae cavae to the right atrium. The **pulmonary circulation** carries deoxygenated blood from the right ventricle through the pulmonary arteries to the lungs and returns oxygenated blood via pulmonary veins to the left atrium. The heart wall consists of three layers: **epicardium** (visceral pericardium), **myocardium** (cardiac muscle, thickest layer), and **endocardium** (inner lining). The cardiac cycle (~0.8 seconds at 75 bpm) includes diastole (filling, ~0.5 s) and systole (contraction/ejection, ~0.3 s). Cardiac output (CO) = Heart Rate (HR) × Stroke Volume (SV); normal CO = 5 L/min. The **Frank-Starling law** states that increased venous return stretches the myocardium, increasing the force of contraction and SV. Electrical conduction follows: SA node (pacemaker, 70–80 bpm) → AV node (delay, allows atrial emptying) → Bundle of His → bundle branches → Purkinje fibers (ventricular activation). The ECG reflects this: P wave (atrial depolarization), QRS complex (ventricular depolarization), T wave (ventricular repolarization). Blood pressure is regulated by baroreceptor reflexes, the renin-angiotensin-aldosterone system (RAAS), and local autoregulatory mechanisms.

Evolution of Heart & Aortic Arches

The vertebrate heart evolved from a simple tubular structure in fish to the four-chambered heart of mammals and birds. In **fish**, the two-chambered heart (one atrium, one ventricle) pumps blood through a single circuit: heart → gills → body → heart. The aortic arch system in fish consists of six pairs of aortic arches connecting the ventral and dorsal aortae. During embryonic development in higher vertebrates, these arches are modified: **Arch I & II** largely regress; **Arch III** forms the common carotid arteries; **Arch IV** — left forms the aortic arch, right forms the proximal right subclavian artery; **Arch V** regresses; **Arch VI** (pulmonary arch) — left forms the ductus arteriosus (ligamentum arteriosum in adults), right forms the right pulmonary artery. In **amphibians**, a three-chambered heart (two atria, one ventricle) allows partial separation of oxygenated and deoxygenated blood with a double circulation (pulmocutaneous and systemic). **Reptiles** have a three-chambered heart with a partially divided ventricle (except crocodilians, which have four chambers). **Birds and mammals** independently evolved a complete four-chambered heart with total separation of pulmonary and systemic circuits, enabling high metabolic rates and endothermy. This evolutionary progression reflects the transition from aquatic to terrestrial life and the increasing demand for efficient oxygen delivery.

Nervous System in Vertebrates

The vertebrate nervous system evolved from a simple nerve net in cnidarians to the highly centralized system of mammals. In **fish**, the brain is dominated by the optic tectum (visual processing) and olfactory bulbs. The spinal cord controls segmental reflexes. In **amphibians**, the forebrain enlarges slightly, and the cerebellum develops for coordinated movement. In **reptiles**, the cerebral cortex begins to develop (dorsal ventricular ridge), and the cerebellum enlarges. In **birds**, the optic lobes remain large, but the basal ganglia (striatum) are highly developed for complex motor patterns (flight, song). In **mammals**, the neocortex (six-layered) expands dramatically, enabling higher cognitive functions, sensory integration, and voluntary motor control. The limbic system (amygdala, hippocampus, hypothalamus) governs emotion and memory. The autonomic nervous system (sympathetic and parasympathetic) regulates visceral functions. The blood-brain barrier (formed by astrocyte end-feet and tight junctions of capillary endothelium) protects the CNS. Comparative neuroanatomy reveals that brain size relative to body size (encephalization quotient) increases through vertebrate evolution, with humans having the highest EQ.

Cranial Nerves

There are **12 pairs of cranial nerves** emerging from the brain and brainstem, designated by Roman numerals I–XII. They are classified as sensory (I, II, VIII), motor (III, IV, VI, XI, XII), or mixed (V, VII, IX, X). The vagus nerve (X) is the most important parasympathetic nerve, innervating thoracic and abdominal viscera.

#	Name	Type	Function	Nucleus / Exit
I	Olfactory	Sensory	Smell (olfaction)	Olfactory epithelium / Cribriform plate
II	Optic	Sensory	Vision	Retina / Optic canal
III	Oculomotor	Motor	Eye movement (4 muscles), pupil constriction, accommodation	Midbrain / Superior orbital fissure
IV	Trochlear	Motor	Superior oblique muscle (eye rotation)	Midbrain / Superior orbital fissure
V	Trigeminal	Mixed	Facial sensation, mastication muscles	Pons / Three divisions: V1, V2, V3
VI	Abducens	Motor	Lateral rectus muscle (eye abduction)	Pons / Superior orbital fissure
VII	Facial	Mixed	Facial expression, taste (anterior 2/3 tongue), salivation	Pons / Internal acoustic meatus
VIII	Vestibulocochlear	Sensory	Hearing and equilibrium	Inner ear / Internal acoustic meatus
IX	Glossopharyngeal	Mixed	Taste (posterior 1/3 tongue), pharynx, parotid gland	Medulla / Jugular foramen
X	Vagus	Mixed	Parasympathetic to thoracic/abdominal viscera, larynx, pharynx	Medulla / Jugular foramen
XI	Accessory	Motor	Sternocleidomastoid and trapezius muscles	Medulla + spinal cord / Jugular foramen
XII	Hypoglossal	Motor	Tongue movements	Medulla / Hypoglossal canal

Sensory Only (I, II, VIII)
Olfactory (smell), Optic (vision), Vestibulocochlear (hearing & balance). Purely afferent — carry information to the CNS.

Motor Only (III, IV, VI, XI, XII)
Oculomotor, Trochlear, Abducens (eye movements), Accessory (neck muscles), Hypoglossal (tongue). Purely efferent.

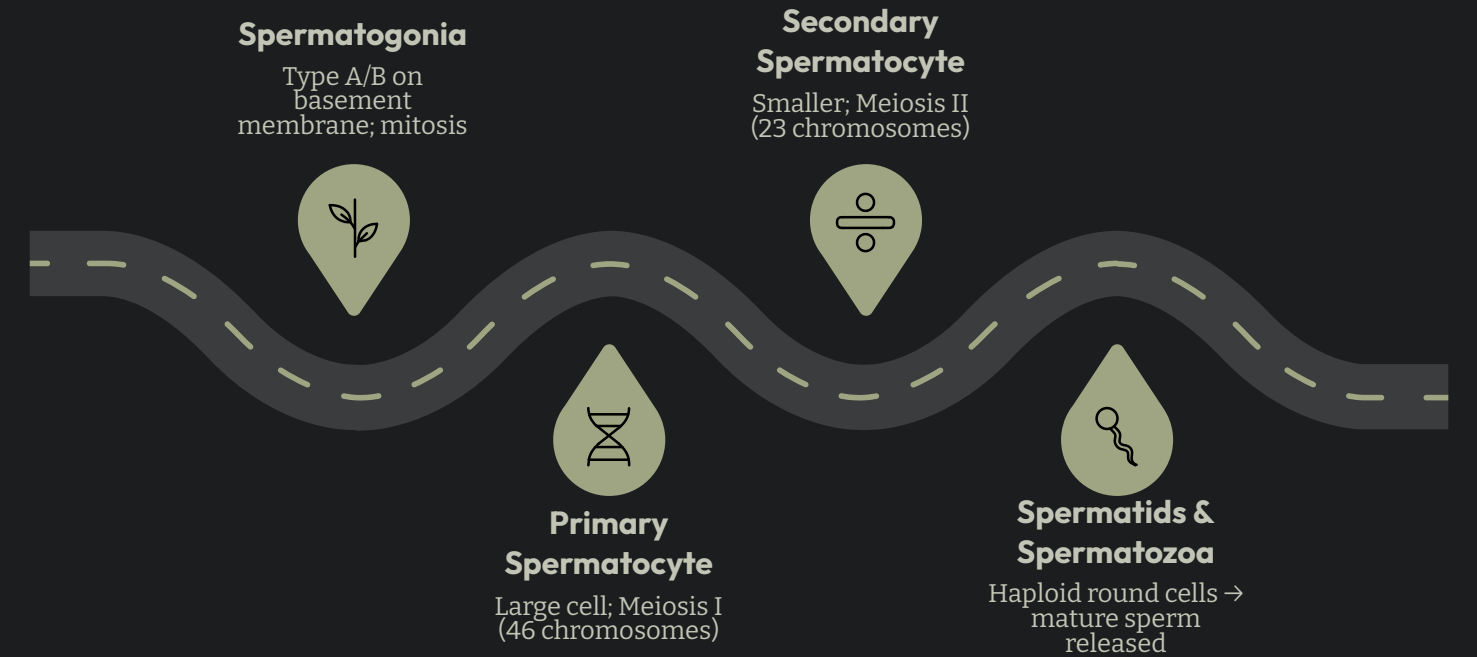
Mixed Nerves (V, VII, IX, X)
Trigeminal, Facial, Glossopharyngeal, Vagus. Carry both sensory and motor fibers. Vagus (X) is the most widespread.

Reproductive Histology: Testis & Ovary

Reproductive histology examines the microscopic structure of the gonads — the testis in males and the ovary in females. These organs are responsible for gametogenesis (spermatogenesis and oogenesis) and steroid hormone production. Understanding their histological organization is essential for comprehending reproductive physiology, endocrinology, and the basis of many reproductive disorders.

Histology of the Testis

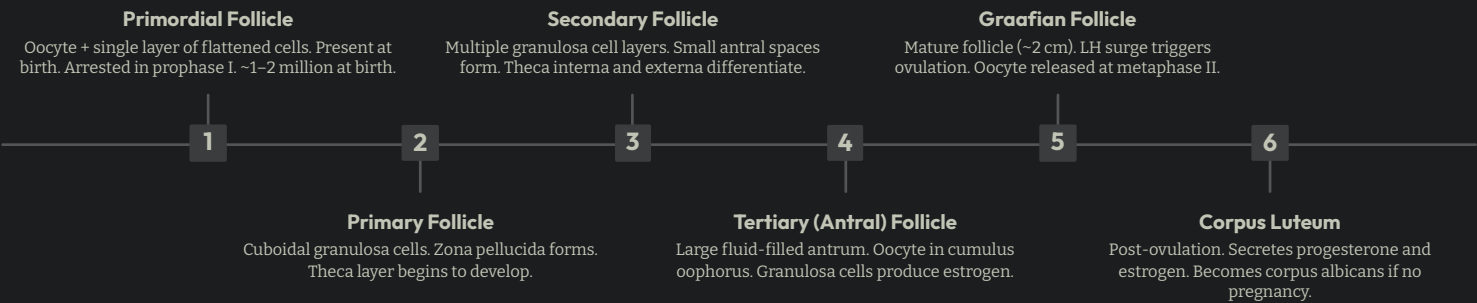
The testis is enclosed by a dense fibrous capsule, the **tunica albuginea**, which thickens posteriorly to form the **mediastinum testis**. From the mediastinum, connective tissue septa divide the testis into approximately 250–300 **lobules**. Each lobule contains 1–4 highly coiled **seminiferous tubules** (total length ~250 m per testis), where spermatogenesis occurs. The seminiferous tubule wall consists of a basement membrane, peritubular myoid cells (contractile, help move sperm), and the **seminiferous epithelium**. The epithelium contains two cell populations: **spermatogenic cells** (in various stages of development) and **Sertoli cells** (sustentacular cells). Sertoli cells extend from the basement membrane to the lumen, forming the **blood-testis barrier** via tight junctions, which protects developing sperm from the immune system and creates a specialized microenvironment. Sertoli cells provide nutritional support, secrete androgen-binding protein (ABP), inhibin, anti-Müllerian hormone (AMH), and phagocytose residual bodies. Spermatogenesis proceeds through several stages: **spermatogonia** (stem cells, Type A = reserve, Type B = differentiating) → **primary spermatocytes** (undergo meiosis I, 46 chromosomes, 4C DNA) → **secondary spermatocytes** (meiosis II, 23 chromosomes, 2C DNA) → **spermatids** (haploid, 23 chromosomes, 1C DNA) → **spermatozoa** (mature sperm, after spermiogenesis). Spermiogenesis involves acrosome formation (from Golgi), flagellum development (from centriole), condensation of nuclear chromatin, and shedding of excess cytoplasm. The entire process takes approximately **64–74 days** in humans. Between seminiferous tubules lie the **interstitial cells of Leydig**, which produce **testosterone** in response to luteinizing hormone (LH). Testosterone is essential for spermatogenesis, secondary sexual characteristics, and anabolic effects. FSH stimulates Sertoli cells to support spermatogenesis. Mature sperm are transported through the **rete testis**, **efferent ductules**, and into the **epididymis**, where they mature and are stored for ~2–3 weeks.



Spermatogenesis is a continuous process beginning at puberty and continuing throughout life. The blood-testis barrier, formed by tight junctions between Sertoli cells, divides the seminiferous epithelium into a basal compartment (containing spermatogonia and early primary spermatocytes) and an adluminal compartment (containing later-stage cells). This barrier is essential for preventing autoimmune reactions against haploid sperm cells, which express antigens not present during immune system development.

Histology of the Ovary

The ovary is covered by a modified peritoneum called the **germinal epithelium** (simple cuboidal) and a dense connective tissue capsule, the **tunica albuginea**. The ovarian stroma is divided into an outer **cortex** (densely cellular, contains ovarian follicles at various stages of development) and an inner **medulla** (loose connective tissue with blood vessels, lymphatics, and nerves). The cortex contains **oocytes** surrounded by follicular cells at various developmental stages. Folliculogenesis begins with **primordial follicles** (oocyte surrounded by a single layer of flattened pregranulosa cells; present at birth, ~1–2 million, arrested in prophase I of meiosis). Under FSH stimulation, primordial follicles develop into **primary follicles** (cuboidal granulosa cells, zona pellucida forms between oocyte and granulosa cells). **Secondary follicles** develop multiple layers of granulosa cells and small fluid-filled spaces (antrum begins to form). **Tertiary (antral) follicles** have a large fluid-filled antrum containing follicular fluid; the oocyte is surrounded by the cumulus oophorus and corona radiata. Theca cells differentiate into theca interna (steroidogenic, produce androgens under LH stimulation) and theca externa (fibrous). Granulosa cells convert androgens to estrogens via aromatase (under FSH stimulation). One follicle becomes the **dominant (Graafian) follicle**, reaching ~2 cm in diameter. At ovulation (triggered by the LH surge), the mature oocyte (arrested in metaphase II) is released with the corona radiata into the peritoneal cavity and captured by the fimbriae of the uterine tube. After ovulation, the collapsed follicle transforms into the **corpus luteum** (granulosa lutein cells and theca lutein cells), which secretes progesterone and estrogen to support the endometrium. If pregnancy does not occur, the corpus luteum degenerates into the **corpus albicans** (fibrous scar) after ~14 days. If pregnancy occurs, hCG from the trophoblast maintains the corpus luteum until the placenta takes over steroidogenesis (~week 10). Unlike spermatogenesis, oogenesis is not continuous — all primary oocytes are formed before birth, and only ~400–500 follicles mature and ovulate during a woman's reproductive lifetime.



Testis: Key Features

- Seminiferous tubules with spermatogenic series
- Sertoli cells: blood-testis barrier, support, ABP, inhibin
- Leydig cells: testosterone production (LH-stimulated)
- Continuous gametogenesis from puberty onward
- ~64–74 days for complete spermatogenesis
- ~200–300 million sperm produced daily

Ovary: Key Features

- Cortex contains follicles at all developmental stages
- Granulosa cells: estrogen production (FSH-stimulated)
- Theca interna: androgen production (LH-stimulated)
- Cyclical gametogenesis (menstrual cycle)
- All oocytes present at birth; ~400–500 ovulated in lifetime
- Corpus luteum: progesterone source post-ovulation

The histological organization of the gonads reflects their dual function: gametogenesis and steroidogenesis. Sertoli cells and granulosa cells are functionally analogous — both support developing gametes, respond to FSH, and produce inhibin. Leydig cells and theca interna cells are similarly analogous — both produce androgens under LH stimulation. This parallel organization underscores the shared embryological origin of testicular and ovarian tissue from the indifferent gonad.